

## RESEARCH PROFILE

Seeking a long-term undergraduate research internship or research-assistant opportunity where I can develop rigorous research habits through implementation, reproduction, experimentation, and technical writing.

### Software Engineering & Developer Tools

Design, requirements, modeling, testing, and code quality.

### AI-Assisted Software Engineering

LLMs and agents for coding, testing, debugging, and workflows.

### Algorithms & Reliable Systems

Retrieval, system modeling, performance, reliability, and security.

## EDUCATION

East China Normal University

2025 - 2029

B.Eng. in Software Engineering, expected 2029

|                   |                                  |                                       |              |
|-------------------|----------------------------------|---------------------------------------|--------------|
| GPA<br>3.81 / 4.0 | MAJOR RANK<br>35 / 253 - TOP 14% | SCHOLARSHIP RANK<br>17 / 253 - TOP 7% | CET-4<br>589 |
|-------------------|----------------------------------|---------------------------------------|--------------|

## SELECTED PROJECTS

ScaffoldMind

May - Jun 2026

Independent Developer / Independent Project

Designed and implemented a local-first workspace for structured computer-science learning, from product flows and interface design through backend APIs and persistence.

- Built guided chat, practice, review, history, learner-profile, and local-settings workflows.
- Implemented React and TypeScript interfaces, Express APIs, SSE streaming, and backend LLM integration.
- Kept API credentials server-side and learning records local.

[React](#) / [TypeScript](#) / [Node.js](#) / [Express](#) / [SSE](#) / [LLM API](#)

Mini-VDB: Dynamic Vector Index & Exact Top-K Retrieval

Jun 2026

Independent Developer / Data Structures and Algorithm Practice / Supervisor: Wang Liping

Implemented the indexing and retrieval core of a small vector database and documented design trade-offs for exact similarity search under dynamic updates.

- Supported insertion, deletion, exact Top-K, and threshold retrieval with Euclidean distance and inner product.
- Implemented open-address hashing, heap-based Top-K selection, KD-tree pruning, and AVX2 sequential scanning.
- Developed eight progressive versions spanning sorting, priority queues, BSTs, and AVL trees.

[C++](#) / [KD-tree](#) / [Top-K Retrieval](#) / [Heap](#) / [AVX2](#)

Intelligent Elevator Collaborative Modeling & Simulation

Jul 2026

Control Modeling Contributor / Five-person course project / Supervisor: Liu Jing

Independently completed the continuous-dynamics and control portion of a collaborative SysML and Simulink elevator model.

- Modeled car, counterweight, load, gravity, friction, braking, and acceleration-to-position dynamics.
- Implemented PID feedback and load compensation with gravity, acceleration, and friction feed-forward terms.
- Validated final position error below 0.01 m for no-load, half-load, and full-load simulations.

[MATLAB](#) / [Simulink](#) / [SysML](#) / [PID Control](#) / [Dynamic Modeling](#)

### ADDITIONAL PROJECT

Maple

Database & Core API Subtask Owner / Four-person project

Owned the persistence and core-data-service subtask for a local mind-map notebook with editing and version-management features.

- Designed node, connection, and metadata schemas with Spring Boot entity mappings.
- Implemented APIs for node CRUD, connection management, and layout-data storage.

[Java 17](#) / [Spring Boot](#) / [JPA](#) / [H2](#) / [SQLite](#) / [REST API](#) / [Git](#)

Jan - Mar 2026

### COMPETITION & DESIGN WORK

Intelligent Water Quality Monitoring & Early-Warning System

Team Lead / Four-person technical proposal / Oct - Nov 2025

Coordinated the submission and contributed to the proposed sensing, acquisition, communication, data-fusion, warning, and energy architecture. Deliverables were a technical proposal, functional diagrams, and device concept design.

Zhiyi

Technical & Functional Proposal Design / Ten-person project / Jun 2026

Authored the technical and functional sections of an education-technology proposal, translating educational goals into software functions, interaction flows, technical architecture, and application scenarios.

### HONORS & AWARDS

Undergraduate Excellence Scholarship

First Class, East China Normal University

2025-2026

Advanced Robotics and Simulation Technology Competition

National Third Prize

Nov 2025

Challenge Cup Entrepreneurship Plan Competition, ECNU Round

Silver Award

Jun 2026

ECNU Coder Freshman Programming Challenge

Third Prize

2025

Boyuan IT Club Owner-Pro

Third Prize

Mar 2026

### TECHNICAL SKILLS

|                 |                                                                         |
|-----------------|-------------------------------------------------------------------------|
| LANGUAGES       | C++, C, Python, Java, TypeScript                                        |
| AI & RETRIEVAL  | LLM application development, prompt engineering, vector search, PyTorch |
| SYSTEMS & TOOLS | Linux, Git/GitHub, unit testing, REST APIs                              |
| MODELING        | MATLAB, Simulink, SysML, dynamic-system modeling                        |

### SELECTED COURSEWORK

|                                                   |                                                  |
|---------------------------------------------------|--------------------------------------------------|
| Data Structures and Algorithms - 4.0/4.0 - 98/100 | Linear Algebra - 4.0/4.0                         |
| Computer Systems - 4.0/4.0 - 90/100               | Mathematics for Information Security - 4.0/4.0   |
| Object-Oriented Programming (C++) - 4.0/4.0       | Mathematical Thinking of AI - 4.0/4.0            |
| Discrete Mathematics - 4.0/4.0                    | Data Structures and Algorithm Practice - 4.0/4.0 |